

Rising Secondary Infertility in India



Source: TH

Why in News?

Secondary **infertility** is emerging as a growing reproductive health concern in India. Analysis of **National Family Health Survey (NFHS)** data shows that its prevalence increased from about **19.5% in the early 1990s to 28.6% by 2015-16** .

- Delayed parenthood, age-related fertility decline, lifestyle disorders, reproductive health complications, and male infertility are among the major factors contributing to the rise.

What is Infertility?

- **Infertility:** It is a disease of the **male or female** reproductive system, defined as the failure to achieve pregnancy after **12 months or more of regular unprotected sexual intercourse** .
 - According to the **World Health Organization (WHO)**, **around one in six people of reproductive age globally experience infertility during their lifetime** .
 - Infertility may result from **male factors, female factors, a combination of both, or unexplained causes** .
- **Primary Infertility:** A pregnancy has never been achieved by an individual or couple.
- **Secondary Infertility:** At least one previous pregnancy has been achieved, but the individual or couple is subsequently unable to conceive again.

What are the Causes of the Rising Trend of Secondary Infertility in India?

- **Demographic Shift towards Delayed Family Completion:** Career consolidation, urban migration, and financial pressures have pushed the first childbirth into the early 30s, leaving a narrower biological window for a second pregnancy.
 - By the time couples plan another child, women are often above **35 years** , when **Diminished Ovarian Reserve (DOR)** and declining egg quality reduce conception chances. Advancing paternal age can also increase **sperm DNA fragmentation**.
- **Post-Partum Anatomical Complications:** Previous pregnancies and deliveries can alter the reproductive environment.

- The rising incidence of **Caesarean sections** can cause pelvic adhesions or **isthmocele (uterine scar defect)** , potentially interfering with implantation and fertility.
- Untreated postpartum infections may also cause **Pelvic Inflammatory Disease (PID)** and fallopian tube blockage.
- **Progression of Reproductive Disorders:** Conditions such as **endometriosis and uterine fibroids** may develop or worsen during the interval between the first and second pregnancy, creating new barriers to conception despite previous fertility.
- **Post-Partum Endocrine and Metabolic Shifts:** Weight gain, sedentary lifestyles, and metabolic changes can disrupt the **Hypothalamic-Pituitary-Ovarian (HPO) axis** , affecting ovulation and reproductive hormones.
- **Growing Burden of PCOS and Thyroid Disorders:** **Polycystic Ovary Syndrome (PCOS)**, **affecting up to 20% of Indian women** , may worsen with age, weight gain, and insulin resistance, leading to chronic anovulation.
 - Postpartum thyroiditis and hypothyroidism can similarly disrupt menstrual cycles and fertility.
- **Overlooked Male Factor:** Weight gain, smoking, alcohol consumption, chronic occupational stress, and metabolic disorders can progressively reduce **sperm concentration, motility, and morphology** after the first child.
- **Growing Exposure to Endocrine-Disrupting Chemicals (EDCs):** Prolonged exposure to pesticides, **microplastics** , and industrial chemicals can interfere with hormonal signalling and adversely affect egg and sperm quality.
- **Urban Air Pollution and Oxidative Stress:** Chronic exposure to **PM2.5 and heavy metals** can generate oxidative stress, which is associated with declining sperm parameters and impaired ovarian function, making environmental pollution an emerging contributor to secondary infertility.

What are the Major Challenges in Addressing Secondary Infertility in India?

- **Legal and Regulatory Bottlenecks:** The current legal framework governing assisted reproduction in India inadvertently penalizes couples facing secondary infertility.
 - **Under the Surrogacy (Regulation) Act, 2021** married couples are explicitly barred from opting for surrogacy if they already have a surviving child (biological, adopted, or via surrogacy). The law only allows an exception if the existing child suffers from a life-threatening disorder or severe mental/physical disability.
 - This can restrict options for couples with medically diagnosed secondary infertility and has raised questions over reproductive autonomy under **Article 21**.
- **Public Health Policy Blind Spots:** Infertility management remains a glaring omission in India's broader public health strategy.
 - Advanced fertility treatments like **In Vitro Fertilization (IVF)** and Intracytoplasmic Sperm Injection (ICSI) are highly capital-intensive, costing upwards of Rs 1.5 to 2 lakhs per cycle.
 - However, these treatments are largely excluded from public health insurance schemes like the **Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)** .
 - India's flagship reproductive health program, the **RMNCH+A (Reproductive, Maternal, Newborn, Child, and Adolescent Health) framework** , heavily prioritizes family planning (contraception) and maternal mortality reduction. Infertility care is largely overlooked as a critical component of reproductive rights.

- **Urban-Private Concentration:** Tertiary reproductive care and **ART (Assisted Reproductive Technology)** clinics are hyper-concentrated in the private sector and limited to **Tier-1 and Tier-2 cities**.
 - Public district hospitals lack the embryology labs and specialized endocrinologists required to treat complex secondary infertility.
- **Socio-Cultural Stigma and Diagnostic Evasion:** The societal approach to secondary infertility actively delays medical intervention.
 - Because the couple already has a child, their grief and medical struggles are frequently minimized or dismissed by society. This lack of social empathy leads to **isolation and prevents couples from seeking timely psychiatric or medical help**.
 - There is a deeply entrenched misconception that a man who has fathered one child is permanently fertile. Consequently, male partners often refuse or **delay undergoing standard semen analysis**.
 - Given that declining sperm quality is a major driver of secondary infertility, this diagnostic evasion causes couples to lose crucial reproductive years.
- **Proliferation of Misinformation and Unregulated Care:** While the ART Act, 2021 attempted to regulate fertility clinics, enforcement remains a challenge.
 - Social media is filled with unregulated clinics promising "100% IVF success in one cycle," preying on the emotional vulnerability of older couples desperate for a second child.
 - Many couples in rural and semi-urban areas waste their critical remaining reproductive years (particularly for women over 35, where ovarian reserve drops rapidly) on unverified **home remedies and alternative treatments before seeking evidence-based allopathic intervention**.

key Initiatives Addressing Infertility in India

- **Assisted Reproductive Technology (Regulation) Act, 2021**
- **Surrogacy (Regulation) Act, 2021**
- **Jiyo Parsi Scheme** (Ministry of Minority Affairs initiative for the Parsi community)
- **Employees' State Insurance Corporation (ESIC) Scheme** (Free IVF coverage for eligible insured women)
- **Central Government Health Scheme (CGHS)** (Partial IVF reimbursement for government employees)
- **State-Specific Free/Subsidized IVF Schemes** (e.g., Goa Free IVF Scheme, Tamil Nadu ART Scheme, Rajasthan ART Policy)

Conclusion

India must adopt a **rights-based approach to address secondary infertility** by reviewing restrictive surrogacy provisions, expanding affordable fertility care under **AB-PMJAY and RMNCH+A**, strengthening infertility services at district hospitals, and combating gender stigma around male infertility. Such reforms can ensure **equitable fertility care and protect reproductive autonomy under Article 21**.

Drishti Mains Question:

The growing burden of secondary infertility exposes the gender bias embedded in India's reproductive healthcare system. Discuss

Frequently Asked Questions (FAQs)

1. What is secondary infertility?

Secondary infertility refers to the inability to conceive or carry a pregnancy after at least one previous pregnancy has been achieved, distinguishing it from primary infertility.

2. What restrictions does the Surrogacy (Regulation) Act, 2021 impose on couples with a surviving child?

The Act generally bars such couples from accessing surrogacy, except where the existing child has specified disabilities or a life-threatening disorder, raising concerns over **reproductive autonomy under Article 21**.

3. What is the significance of the Assisted Reproductive Technology (Regulation) Act, 2021?

The Act regulates **ART clinics and banks**, prescribes standards and registration requirements, and seeks to prevent unethical practices and protect women and children born through ART procedures.

4. Why is infertility care considered a public health policy gap in India?

Advanced fertility treatments such as **IVF and ICSI remain largely outside public insurance coverage**, while specialised ART services are concentrated in private facilities and major cities.

5. What measures can improve access to fertility care in India?

India should integrate infertility care into **UHC, AB-PMJAY, and RMNCH+A**, strengthen district-level fertility services, regulate misleading ART marketing, promote male fertility evaluation, and protect reproductive autonomy under **Article 21**.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/w1rWq6Kslkc](https://www.youtube.com/embed/w1rWq6Kslkc)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. In the context of recent advances in human reproductive technology, “Pronuclear Transfer” is used for (2020)

- (a) fertilization of egg in vitro by the donor sperm
- (b) genetic modification of sperm producing cells
- (c) development of stem cells into functional embryos
- (d) prevention of mitochondrial diseases in offspring

Ans: (d)

Q. Consider the following statements: (2020)

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.
2. A person’s genome can be edited before birth at the early embryonic stage.
3. Human induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 2 only
- (d) 1, 2 and 3

Ans: (d)

Mains

Q1. “Empowering women is the key to control the population growth.” Discuss. **(2019)**

Q2. Critically examine the effect of globalization on the aged population in India. **(2013)**

Q3. Discuss the main objectives of Population Education and point out the measures to achieve them in India in detail. **(2021)**

Sub-Mission on Agricultural Mechanization



Source: PIB

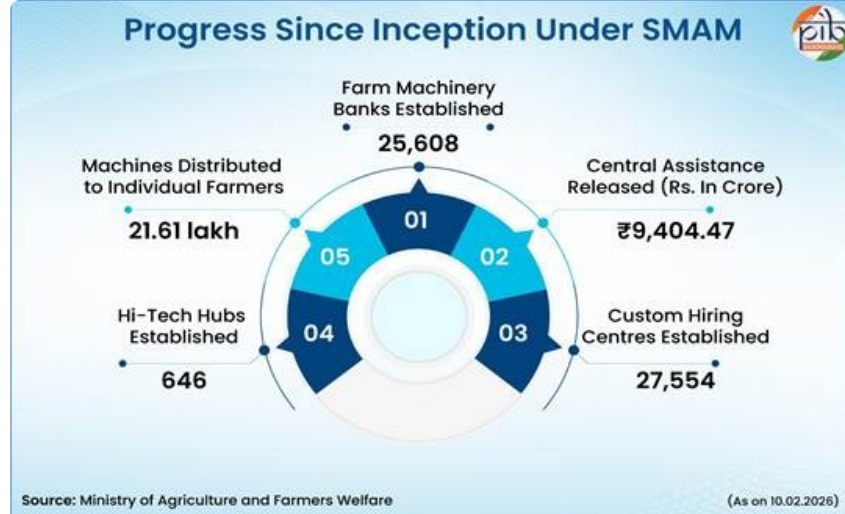
Why in News?

The Sub-Mission on Agricultural Mechanization (SMAM) has **expanded access to farm machinery and modern technologies** for small farmers, women and disadvantaged groups, with special focus on underserved and North-Eastern States.

- Since its inception, ₹9,404.47 crore has supported the distribution of 21.61 lakh machines, while ₹52.5 crore has enabled 40,928 drone demonstrations across 40,918 hectares to **promote precision agriculture**.

What is the Sub-Mission on Agricultural Mechanization?

- **About:** The [Sub-Mission on Agricultural Mechanization \(SMAM\)](#) was launched in **2014-15 as a Centrally Sponsored Scheme** under the [Rashtriya Krishi Vikas Yojana \(RKVY\)](#) to promote farm mechanization across India.
- **Objective:** It follows the principle of “**reach the unreached**” by extending mechanization benefits to **small and marginal farmers, women, Scheduled Castes, Scheduled Tribes, Farmer-Producer Organizations , Self-Help Groups and rural entrepreneurs** .
- **Custom Hiring Centres:** SMAM supports the **establishment of Custom Hiring Centres** comprising farm machinery, implements and equipment that farmers can rent at affordable rates.
- **Machinery and Technology Support:** The scheme **promotes the distribution of farm machinery and the creation of hubs for hi-tech** and high-value agricultural equipment.
- **Regional and Structural Focus:** Special attention is given to **regions with low farm power availability**, while rental services help address constraints such as small landholdings and the high cost of purchasing machinery.
- **Financial Support:** From **2014-15 to 2025-26, central assistance of ₹9,404.47 crore** supported the distribution of **21.61 lakh machines to individual farmers**, along with the establishment of 27,554 Custom Hiring Centres, 646 Hi-tech Hubs and 25,608 Farm Machinery Banks.
- **Expansion:** The number of beneficiaries receiving support for individual farm machinery ownership **increased from 2.07 lakh in 2020-21 to 2.32 lakh in 2024-25**, indicating wider scheme coverage.
- **Funding Pattern:** The Centre and States share expenditure in a **60:40 ratio for most States and 90:10 for North-Eastern and Himalayan States, while Union Territories receive 100 percent** Central funding.
- **Capacity Building:** Demonstrations, **training programmes and targeted Information, Education and Communication activities** are undertaken to improve awareness, skills and adoption of farm mechanization.
- **Testing and Certification:** SMAM also supports the performance testing and certification of agricultural machinery to ensure its quality, efficiency and suitability for farmers.



Rashtriya Krishi Vikas Yojana

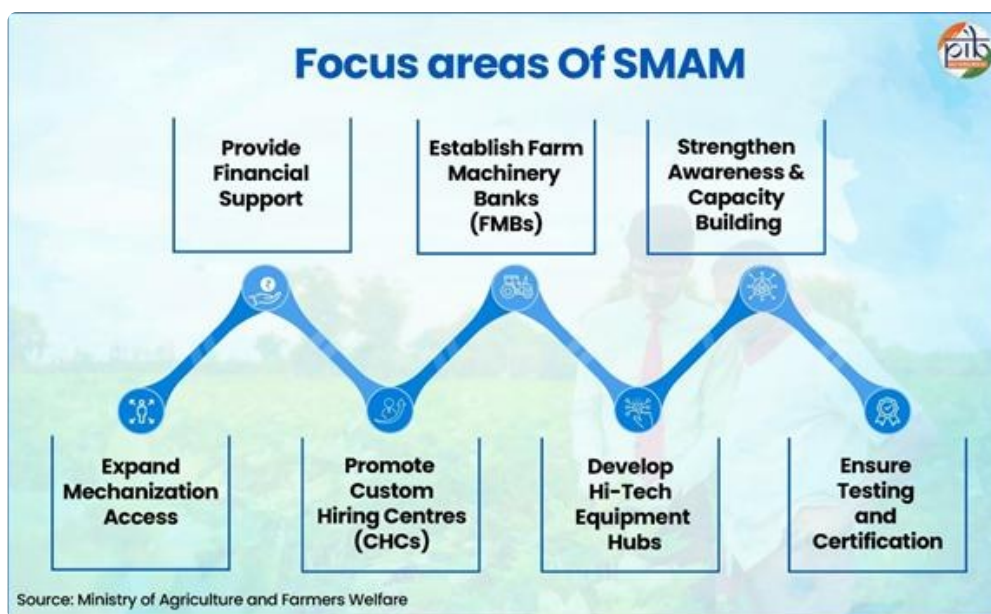
- **About:** The Rashtriya Krishi Vikas Yojana (RKVY) , **launched in 2007, is a flagship scheme** aimed at encouraging States to increase public investment in agriculture and allied sectors.
- **State Flexibility:** It gives States autonomy to design and implement projects according to their local agricultural priorities and requirements.
- **Coverage:** The scheme supports interventions such as **crop production, agricultural infrastructure** , farm mechanization and value addition.
- **RKVY-RAFTAR:** In 2017-18, RKVY was restructured as RKVY-RAFTAR—Remunerative Approaches for Agriculture and Allied Sector Rejuvenation—with greater emphasis on **strengthening pre-harvest and post-harvest infrastructure**.

What are the Strategic Pillars of SMAM for Enhancing Farm Productivity?

- **Promotion of Agricultural Mechanization through Training, Testing, Demonstration and Post-Harvest Mechanization:** Improves farm productivity and efficiency by expanding access to modern agricultural technologies. It also strengthens post-harvest infrastructure for processing, storage, value addition and crop-residue management.
- **Financial Assistance for the Procurement of Agricultural Machinery and Equipment:** Provides subsidies through the **Direct Benefit Transfer (DBT)** system for individual ownership of agricultural machinery. It covers 40% of the machinery cost for general beneficiaries and **50% for SC/ST farmers, small and marginal farmers and beneficiaries in North-Eastern States** . Small and marginal farmers also receive **₹2,000 per hectare for mechanized services**, including drone services, through CHCs, SHGs and FPOs.
- **Establishment of Farm Machinery Banks for Custom Hiring:** Provides **80-90% financial assistance to SHGs, FPOs and local institutions** for acquiring agricultural machinery. Custom Hiring Centres receive assistance equivalent to **40% of the project cost for projects costing up to ₹250 lakh** , while **Farm Machinery Banks** receive 80% assistance for projects costing up to ₹30 lakh.
- **Establishment of Hi-Tech and High-Productivity Equipment Hubs:** Supports hubs equipped with advanced, high-capacity and crop-specific machinery to improve operational efficiency and expand

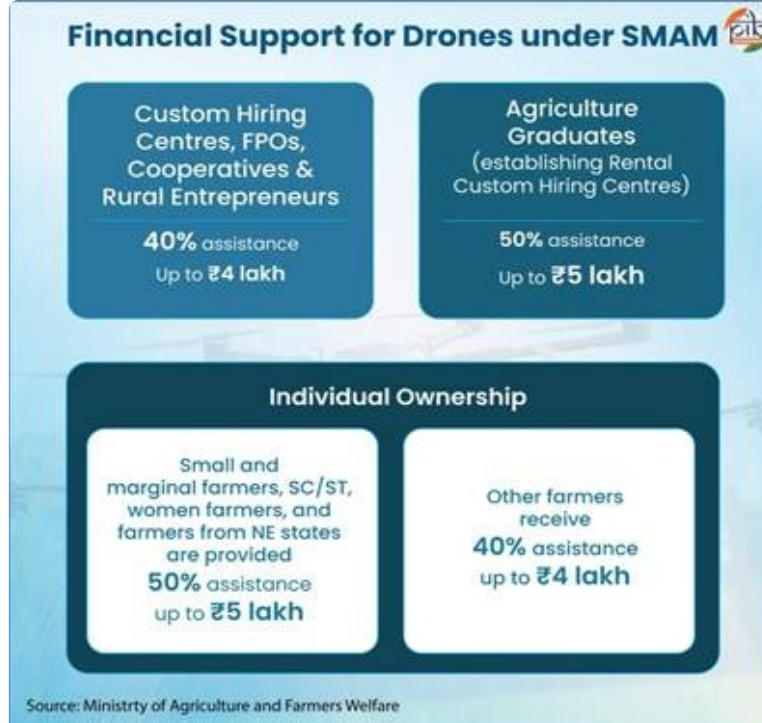
access to high-value agricultural equipment.

- **Promotion of Agricultural Mechanization in the North-Eastern Region:** Provides enhanced assistance through region-specific interventions, including subsidies of up to **100% for small machinery and 95% support for Farm Machinery Banks.**



What is the Role of Farm Mechanization and Drone Technology under SMAM?

- **Role of Farm Mechanization:** Farm mechanization uses **machinery, equipment and modern technologies** across the crop-production cycle, including land preparation, sowing, irrigation, plant protection, harvesting and post-harvest management.
- **Benefits:** It reduces dependence on manual labour and animal power, **enables timely and efficient agricultural operations**, lowers production costs and improves the quality and productivity of farming practices.
- **Promotion of Drone-Based Farming:** The Sub-Mission on Agricultural Mechanization **encourages the use of drones for agricultural operations**. Under the scheme, ₹52.50 crore was provided to the [Indian Council of Agricultural Research](#) for promoting drone adoption through field demonstrations.
- **Kisan Drone Demonstrations:** From **2023-24 to 2025-26**, **ICAR, State Agricultural Universities and Krishi Vigyan Kendras** conducted 40,928 Kisan Drone demonstrations covering 40,918 hectares. These focused on applying nutrients, fertilizers and agrochemicals according to prescribed standard operating procedures.
- **Financial Assistance:** ICAR institutes, KVKs and State Agricultural Universities receive **100% assistance of up to ₹10 lakh per drone for procurement and demonstrations**. Farmer-Producer Organizations receive grants of up to 75%, while agencies providing drone-based services receive contingency support of ₹6,000 per hectare.
- **Support for Women Farmers:** SMAM **earmarks 30% of its total funds for women farmers to improve their access to agricultural machinery** and strengthen their participation in mechanized farming.



Conclusion

SMAM has improved the reach and inclusiveness of farm mechanization, particularly for small farmers, women and disadvantaged groups. However, its long-term impact will depend on **affordable access, effective training**, region-specific implementation and the adoption of sustainable, climate-smart technologies.

Drishti Mains Question

Farm mechanization can enhance agricultural productivity, but its benefits often remain limited for small and marginal farmers. Discuss.

Frequently Asked Questions

1. What is the Sub-Mission on Agricultural Mechanization?

The Sub-Mission on Agricultural Mechanization is a centrally sponsored scheme launched in 2014-15 to promote access to farm machinery, Custom Hiring Centres, Farm Machinery Banks and modern agricultural technologies.

2. Which scheme includes SMAM as a component?

SMAM is implemented as a component of the Rashtriya Krishi Vikas Yojana.

3. Why are Custom Hiring Centres important?

Custom Hiring Centres allow farmers, especially small and marginal farmers, to rent expensive farm machinery at affordable rates.

4. How does SMAM support drone-based farming?

SMAM provides financial support for drone demonstrations and drone procurement by institutions and Farmer-Producer Organizations.

5. Why is SMAM important for inclusive agriculture?

It focuses on small and marginal farmers, women farmers, SC/ST communities, FPOs, SHGs and underserved regions, making mechanization more accessible and inclusive.

UPSC Civil Services Examination, Previous Year Questions

Prelims

Q. Which one of the following best describes the concept of ‘Small Farmer Large Field’? [2023]

- a) Resettlement of a large number of people, uprooted from their countries due to war, by giving them a large cultivable land which they cultivate collectively and share the produce.
- b) Many marginal farmers in an area organize themselves into groups and synchronize and harmonize selected agricultural operations.
- c) Many marginal farmers in an area together make a contract with a corporate body and surrender their land to the corporate body for a fixed term for which the corporate body makes a payment of agreed amount to the farmers.
- d) A company extends loans, technical knowledge and material inputs to a number of small farmers in an area so that they produce the agricultural commodity required by the company for its manufacturing process and commercial production.

Ans: (b)

Mains

Q. What are the present challenges before crop diversification? How do emerging technologies provide an opportunity for crop diversification? (2021)

Q. What are the major factors responsible for making rice-wheat system a success? In spite of this success, how has this system become bane in India? (2020)

Q. How has the emphasis on certain crops brought about changes in cropping patterns in recent past? Elaborate the emphasis on millets production and consumption. (2018)

1st UN Global Dialogue on AI Governance



Source: ET

Recently, the inaugural **UN Global Dialogue on AI Governance** was held in Geneva, bringing together governments, industry, academia and civil society to develop a common approach to governing **artificial intelligence**. A second session will be held in New York in May 2027.

- **About:** The Dialogue was established under **UN General Assembly Resolution 79/325** following the **Global Digital Compact adopted in 2024**. It aims to **strengthen international cooperation, scientific understanding, accountability and capacity-building** in AI governance.
- **India's Position:** India called for a **human-centric, inclusive, safe, secure and trustworthy global AI framework** in which technological development remains subject to meaningful human oversight.
- **Foundational Principles:** India argued that **responsible AI governance must protect human rights, prevent misuse and discrimination**, maintain accountability and ensure that humans retain final authority in critical areas such as healthcare, policing and justice.
- **Global South and Capacity Gaps:** India stressed that **developing countries must receive technological, financial and institutional support** to participate meaningfully in global AI rule-making and benefit equitably from AI-led development.
- **India's Domestic Approach:** India's **National Strategy for AI and National AI Governance Guidelines** follow the philosophy of **"AI for All,"** seeking to promote inclusive growth and innovation without imposing restrictions that unnecessarily hinder technological progress.
- **Major Global Concerns:** The Dialogue highlighted risks including **deepfakes, sexual abuse, threats to children, autonomous weapons, deceptive AI systems, environmental costs, unequal access** and the possibility that AI could deepen existing economic, security and sovereignty gaps.
- **Significance:** India and the UN called for **consensus-based multilateral governance, harmonised safety testing, clear legal responsibility, transparent environmental reporting, renewable-powered data centres and stronger protection for children and human rights** so that AI becomes an instrument of equitable progress rather than a multiplier of inequality.

Read more: [Transforming Governance with AI and DPI](#)